Riemannian Geometry

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1 Lecture 2

Tensor Fields and $C^\infty(M)$-Multilinear Maps

Setup: Consider a tensor field $F \in \mathcal{T}_l^k M$. Let $\omega^1, \ldots, \omega^k \in \Omega^1(M)$ be 1-forms and $X_1, \ldots, X_l \in \mathfrak{X}(M)$ be vector fields.

We can define a map:

$$\mathcal{F} : \underbrace{\Omega^1(M) \times \cdots \times \Omega^1(M)}_{k} \times \underbrace{\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M)}_{l} \to C^\infty(M)$$

by the pointwise evaluation:

$$\mathcal{F}(\omega^1, \ldots, \omega^k, X_1, \ldots, X_l)(p) = F(\omega^1(p), \ldots, \omega^k(p), X_1(p), \ldots, X_l(p))$$

Clearly, $\mathcal{F}$ is $C^\infty(M)$-multilinear.

Lemma 1.1. *Any $C^\infty(M)$-multilinear map $\mathcal{F}$ comes from a unique tensor field $F \in \mathcal{T}_l^k M$.*

- **Algebraic Result:** For finite-dimensional vector spaces, we have the isomorphism $(V \otimes \cdots \otimes V)^* \cong V^* \otimes \cdots \otimes V^*$. In the context of manifolds, this extends to the module level over $C^\infty(M)$.

1.1 Flow of a Vector Field

Definition 1.2. Let $X \in \mathfrak{X}(M)$ be a vector field. The *flow* of X is a collection of (local) diffeomorphisms $\{\Phi_t\}$ such that if $\gamma(t) = \Phi_t(p)$, then:

$$\frac{d\gamma}{dt} = X(\gamma(t)), \quad \gamma(0) = p$$

Definition 1.3. X is called *complete* if the flow $\Phi_t(p)$ exists for all $p \in M$ and all $t \in \mathbb{R}$. In this case, $\{\Phi_t\}$ forms a 1-parameter group of diffeomorphisms: $\Phi_{s+t} = \Phi_s \circ \Phi_t$.

Proposition 1.4. *If M is a compact manifold, then every vector field X on M is complete.*

(Note: From now on, we assume vector fields are complete for simplicity.)

1.2 The Lie Derivative

Lie Derivative of Functions

Definition 1.5. For $f \in C^\infty(M)$, the Lie derivative of f along X is:

$$\mathcal{L}_X f = \left.\frac{d}{dt}\right|_{t=0} (\Phi_t^* f) = \left.\frac{d}{dt}\right|_{t=0} (f \circ \Phi_t) = Xf$$

Lie Derivative of Differential Forms

Definition 1.6. For $\omega \in \Omega^k(M)$, the Lie derivative is:

$$\mathcal{L}_X \omega = \left.\frac{d}{dt}\right|_{t=0} \Phi_t^* \omega$$

Lemma 1.7. *The Lie derivative commutes with the exterior derivative: $\mathcal{L}_X(d\omega) = d(\mathcal{L}_X \omega)$.*

Proof. Since the pull-back Φ_t^* commutes with d:

$$\mathcal{L}_X(d\omega) = \left.\frac{d}{dt}\right|_{t=0} \Phi_t^*(d\omega) = \left.\frac{d}{dt}\right|_{t=0} d(\Phi_t^* \omega) = d\left(\left.\frac{d}{dt}\right|_{t=0} \Phi_t^* \omega\right) = d(\mathcal{L}_X \omega)$$

□

Interior Multiplication

Define the operator $i_X : \Omega^k(M) \to \Omega^{k-1}(M)$ by:

$$(i_X\omega)(X_1, \ldots, X_{k-1}) = \omega(X, X_1, \ldots, X_{k-1})$$

i_X is a graded derivation of degree -1:

$$i_X(\omega \wedge \eta) = (i_X\omega) \wedge \eta + (-1)^{\deg \omega}\omega \wedge (i_X\eta)$$

1.3 Cartan's Magic Formula

Lemma 1.8. *The algebra $\Omega^*(M)$ is generated by $C^\infty(M)$ and $\{df \mid f \in C^\infty(M)\}$.*

Proof. Choose a coordinate neighborhood (U, x^i). Any $\omega \in \Omega^k(U)$ can be written as $\omega = \sum \omega_I dx^I$. Using a partition of unity $\sum \phi_i = 1$ subordinate to an atlas $\{U_i\}$, we can write any global form as a sum of such local terms. □

Proposition 1.9 (Cartan's Magic Formula)**.** $\mathcal{L}_X = di_X + i_X d$

Proof. Both $\mathcal{L}_X$ and $di_X + i_X d$ are graded derivations of degree 0 on $\Omega^*(M)$. Thus, it suffices to check the identity on generators:

1. **On functions $f \in C^\infty(M)$:**

$$(di_X + i_X d)f = d(i_X f) + i_X(df) = 0 + df(X) = Xf = \mathcal{L}_X f$$

2. **On exact 1-forms df:**

$$(di_X + i_X d)df = d(i_X df) + i_X(d^2 f) = d(Xf) + 0 = d(\mathcal{L}_X f) = \mathcal{L}_X(df)$$

Since the operators agree on generators and satisfy the same Leibniz rule, they are equal. □

1.4 Lie Derivative of Vector Fields

Definition 1.10. Let $Y \in \mathfrak{X}(M)$. Its Lie derivative along X is defined as:

$$\mathcal{L}_X Y = \frac{d}{dt}\bigg|_{t=0} (\Phi_{-t})_* Y = \frac{d}{dt}\bigg|_{t=0} (\Phi_t^* Y)$$

Proposition 1.11. *Let ω be a 1-form and X, Y be vector fields. Then:*

$$X(\omega(Y)) = (\mathcal{L}_X\omega)(Y) + \omega(\mathcal{L}_X Y)$$

Proof. Evaluating at $p \in M$:

$$\begin{aligned}
X(\omega(Y))_p &= \frac{d}{dt}\bigg|_{t=0} (\omega(Y))(\Phi_t(p)) \\
&= \frac{d}{dt}\bigg|_{t=0} \langle \Phi_t^*\omega, \Phi_t^* Y\rangle_p \\
&= \langle \frac{d}{dt}\bigg|_{t=0} \Phi_t^*\omega, Y\rangle_p + \langle \omega, \frac{d}{dt}\bigg|_{t=0} \Phi_t^* Y\rangle_p \\
&= (\mathcal{L}_X\omega)(Y) + \omega(\mathcal{L}_X Y)
\end{aligned}$$

□

Proposition 1.12. *$\mathcal{L}_X Y = [X, Y]$, where $[X, Y]$ is the Lie bracket.*

Proof. Let $f \in C^\infty(M)$ be a test function. Consider the 1-form df:

$$\begin{aligned}
\langle df, \mathcal{L}_X Y \rangle &= X(df(Y)) - (\mathcal{L}_X df)(Y) \quad \text{(using the previous Prop)} \\
&= X(Yf) - d(Xf)(Y) \\
&= X(Yf) - Y(Xf) = [X, Y]f
\end{aligned}$$

Since this holds for all f, $\mathcal{L}_X Y = [X, Y]$. $\square$

2 Lecture 3

2.1 Riemannian Metric

Definition 2.1. A **Riemannian metric** on a smooth manifold M is a covariant tensor field g of type $(0,2)$ which is symmetric and positive definite. Specifically, for each $p \in M$, $g_p : T_pM \times T_pM \to \mathbb{R}$ satisfies:

1. **Symmetry:** $g(X,Y) = g(Y,X)$ for all $X, Y \in \mathfrak{X}(M)$.
2. **Positive Definiteness:** $g(X,X) \geq 0$, and $g(X,X) = 0 \iff X = 0$.

Notation: We often write $\langle X, Y \rangle = g(X,Y)$.

Proposition 2.2. *Every smooth manifold M admits a Riemannian metric.*

Sketch of Proof. Take an atlas $\{U_i\}$ of M. On each U_i, define a local metric $g_i = \sum (dx^k)^2$. Let $\{\phi_i\}$ be a partition of unity subordinate to $\{U_i\}$. Then $g = \sum \phi_i g_i$ is a well-defined global Riemannian metric because a convex combination of positive definite symmetric forms remains positive definite and symmetric. □

Suppose we are given a local coordinate $(U, \{x^1, ..., x^n\})$, then we can expressed the metric locally as $g = g_{ij} dx^i dx^j$ since g is symmetric. Further, if M has a local frame $\{E_1, ..., E_n\}$ with coframe $\{\varepsilon^1, ..., \varepsilon^n\}$, then $g = \sum g(E_i, E_j)\varepsilon^i \varepsilon^j$. We can consider the case where $\{E_i\}$ is orthonormal, in this case, $g = \sum (\varepsilon^i)^2$.

Proposition 2.3 (Existence of Orthonormal Frame)**.** *If $\{X_j\}$ is a local frame on (M,g), then there exists an orthonormal frame $\{E_i\}$ obtained from $\{X_j\}$ via the Gram-Schmidt process.*

2.2 Isometries

Definition 2.4. Let (M,g) and $(\tilde{M}, \tilde{g})$ be Riemannian manifolds.

1. A diffeomorphism $\Phi : M \to \tilde{M}$ is an **isometry** if $g = \Phi^* \tilde{g}$, i.e.,
$$g_p(X,Y) = \tilde{g}_{\Phi(p)}(d\Phi_p(X), d\Phi_p(Y)) \quad \forall X, Y \in T_pM$$
2. The set of all isometries from (M,g) to itself forms the **Isometry Group**, denoted $\mathrm{Isom}(M,g)$.

2.3 Examples of Constructions

1. Induced Metric: If $i : M \to \tilde{M}$ is an immersion and $\tilde{g}$ is a metric on $\tilde{M}$, then $g = i^* \tilde{g}$ is the induced Riemannian metric on M.
2. Warped Product: Given $(M_1, g_1), (M_2, g_2)$ and $f \in C^\infty(M_1, \mathbb{R}^+)$, the warped product $M_1 \times_f M_2$ has the metric:
$$g = g_1 + f^2 g_2$$
Formally: $g((X_1, Y_1), (X_2, Y_2)) = g_1(X_1, X_2) + f^2(p_1) g_2(Y_1, Y_2)$.

Example 2.5 (Warped Product)**.** Here are some examples of warped products:

- The polar coordinates on $\mathbb{R}^2 \setminus \{0\}$ can be viewed as a warped product $(0, \infty) \times_r S^1$ with metric $dr^2 + r^2 d\theta^2$.
- The sphere $S^n \setminus \{\text{north, south poles}\}$ can be represented as $(0, \pi) \times_{\sin r} S^{n-1}$ with metric $dr^2 + \sin^2(r) g_{S^{n-1}}$.

Riemann Submanifolds

Lemma 2.6. *Let $i : M \to \tilde{M}$ be an embedding and $\tilde{g}$ a Riemannian metric on $\tilde{M}$. Then the induced metric $g = i^*\tilde{g}$ makes (M, g) a Riemannian submanifold of $(\tilde{M}, \tilde{g})$.*

Riemann Products

Given two Riemannian manifolds (M_1, g_1) and (M_2, g_2), the product manifold $M_1 \times M_2$ has a natural Riemannian metric defined by:

$$g((X_1, Y_1), (X_2, Y_2)) = g_1(X_1, X_2) + g_2(Y_1, Y_2)$$

2.4 Vector Bundles and Distributions

Let $E \to M$ be a vector bundle. An **inner product on** E is a smooth choice of inner product $\langle \cdot, \cdot \rangle$ on each fiber E_m. For any smooth sections $s_1, s_2 \in \Gamma(E)$, the function $m \mapsto \langle s_1(m), s_2(m) \rangle$ is in $C^\infty(M)$.

Vertical and Horizontal Bundles: Let $\pi : \tilde{M} \to M$ be a submersion.

- **Vertical Bundle:** $T^V\tilde{M} = \ker(d\pi)$.
- **Splitting:** A choice of **Horizontal Distribution** $T^H\tilde{M}$ such that $T\tilde{M} = T^V\tilde{M} \oplus T^H\tilde{M}$.
- **Riemannian Choice:** If $\tilde{M}$ has metric $\tilde{g}$, we naturally define $T^H\tilde{M} = (T^V\tilde{M})^\perp$.

Proposition 2.7. *Let $\pi : \tilde{M} \to M$ be smooth submersion of Riemannian manifolds. Then*

- *Every smooth vector field W on $\tilde{M}$ can be uniquely decomposed as $W = W^V + W^H$ with $W^V \in T^V\tilde{M}$ and $W^H \in T^H\tilde{M}$.*
- *For any smooth vector field X on M, there exists a unique horizontal lift $\tilde{X}$ on $\tilde{M}$ such that $d\pi(\tilde{X}) = X$.*

2.5 Riemannian Submersion

Definition 2.8. A submersion $\pi : (\tilde{M}, \tilde{g}) \to (M, g)$ is a **Riemannian submersion** if for all $\tilde{p} \in \tilde{M}$, the map $d\pi_{\tilde{p}}$ restricted to the horizontal space is a linear isometry:

$$d\pi_{\tilde{p}} : (T^H_{\tilde{p}}\tilde{M}, \tilde{g}) \xrightarrow{\cong} (T_{\pi(\tilde{p})}M, g)$$

Recall that we say a group acts **freely** on X if $gX = X$ implies $g = e$. We say the action is **proper** if the map $G \times \tilde{M} \to \tilde{M} \times \tilde{M}$, $(g, p) \mapsto (gp, p)$ is a proper map (i.e. the preimage of any compact set is compact).

Proposition 2.9. *Let G be a Lie group acting freely and properly on $(\tilde{M}, \tilde{g})$ by isometries. Then the quotient $M = \tilde{M}/G$ is a smooth manifold and admits a unique Riemannian metric g such that the projection $\pi : \tilde{M} \to M$ is a Riemannian submersion.*

Proof. First, what is $\tilde{M}/G$? This is the set of orbits and the topology is the quotient topology. We define the Riemann metric on M as follows: For $p \in M$, choose representative $\tilde{p} \in \tilde{M}$ with $\pi(\tilde{p}) = p$. For vectors u, $v \in T_pM$, define

$$g(u, v) = \tilde{g}(\tilde{u}, \tilde{v}) \quad \tilde{u},\ \tilde{v} \in T^H_{\tilde{p}}\tilde{M} \text{ so that } d\pi_{\tilde{p}}(\tilde{u}) = u \text{ and } d\pi_{\tilde{p}}(\tilde{v}) = v.$$

We have to check this is well-defined. If we choose another representative $\tilde{p}' = g\tilde{p}$ for some $g \in G$, then since G acts by isometries, we have

$$\tilde{g}((dg)_{\tilde{p}}(\tilde{u}), (dg)_{\tilde{p}}(\tilde{v})) = \tilde{g}(\tilde{u}, \tilde{v}).$$

Also, since $\pi \circ g = \pi$, we have $(dg)_{\tilde{p}}$ maps horizontal spaces to horizontal spaces because it is an isometry. Thus the definition of g is independent of the choice of representative.

Finally, smoothness of g follows from the smoothness of $\tilde{g}$ and the local triviality of the bundle structure induced by the free and proper action. $\square$

2.6 Riemann Covering Spaces

Definition 2.10. Let $\pi : \tilde{M} \to M$ be a smooth map. We say π is a **covering map** if for every $m \in M$, there exists an open neighborhood $U \subset M$ of m such that:

$$\pi^{-1}(U) = \coprod_i V_i$$

where each $V_i \subset \tilde{M}$ is an open set, and the restriction $\pi|_{V_i} : V_i \to U$ is a **diffeomorphism** for every i.

Metrics on Covering Spaces

If g is a Riemannian metric on the base manifold M, we can naturally obtain a Riemannian metric $\tilde{g}$ on the covering space $\tilde{M}$ via the pullback map:

$$\tilde{g} = \pi^* g$$

Since π is a local diffeomorphism, $\tilde{g}$ is a well-defined Riemannian metric on $\tilde{M}$, making π a local isometry.

*Normal Coverings and Descent of Metrics

Now consider the inverse problem: given a Riemannian metric $\tilde{g}$ on $\tilde{M}$, when does it descend to a metric g on M such that $\pi^* g = \tilde{g}$?

Definition 2.11. $\pi : \tilde{M} \to M$ is a **normal (or Galois) covering** if the fundamental group $\pi_1(\tilde{M})$ is a normal subgroup of $\pi_1(M)$. In this case, the group of Deck transformations is:

$$G = \mathrm{Deck}(\tilde{M}/M) \cong \pi_1(M)/\pi_1(\tilde{M})$$

Proposition 2.12. *Suppose $\pi : \tilde{M} \to M$ is a normal covering and $\tilde{g}$ is a Riemannian metric on $\tilde{M}$. There exists a unique Riemannian metric g on M such that $\pi^* g = \tilde{g}$ if and only if the group G acts on $(\tilde{M}, \tilde{g})$ by **isometries**.*

3 Lecture 4

3.1 Raising and Lowering Indices (Musical Isomorphisms)

The Riemannian metric g provides a natural isomorphism between the tangent bundle TM and the cotangent bundle T^*M.

Lowering Indices (The Flat Operator $\flat$)

Let $X \in T_pM$. We define $X^\flat \in T_p^*M$ by:

$$X^\flat(Y) = \langle X, Y \rangle_g = g(X, Y)$$

In local coordinates, if $X = X^j \partial_j$, then:

$$X^\flat(\partial_i) = g(\partial_j, \partial_i) X^j = g_{ij} X^j$$

Thus, the component expression is:

$$X^\flat = (g_{ij} X^j) dx^i = X_i dx^i$$

Raising Indices (The Sharp Operator $\sharp$)

Let $\omega \in T_p^*M$. We define $\omega^\sharp \in T_pM$ by the condition:

$$\langle \omega^\sharp, X \rangle_g = \omega(X) \quad \forall X \in T_pM$$

In local coordinates, if $\omega = \omega_i dx^i$, then:

$$\omega^\sharp = (g^{ij} \omega_i) \partial_j = \omega^j \partial_j$$

where (g^{ij}) is the inverse matrix of (g_{ij}), i.e., $g^{ij} g_{jk} = \delta^i_k$.

Examples

1. **Gradient:** For $f \in C^\infty(M)$, the differential $df = \partial_i f dx^i$ is a 1-form. Its "sharp" version is the gradient vector field:
$$\text{grad} f = (df)^\sharp = g^{ij} \partial_i f \partial_j$$
2. **Inner Product of 1-forms:** Let $\omega, \eta \in \Omega^1(M)$. Their inner product is defined via the isomorphism:
$$\langle \omega, \eta \rangle_g = \langle \omega^\sharp, \eta^\sharp \rangle_g = g^{ij} \omega_i \eta_j$$

3.2 Volume Form and Density

Let (M, g) be an oriented Riemannian manifold of dimension n.

Definition 3.1. The **Riemannian volume form** dV is the unique n-form such that for any oriented orthonormal basis $\{E_1, \ldots, E_n\}$ of T_pM, we have:

$$dV(E_1, \ldots, E_n) = 1$$

Proposition 3.2. *In local coordinates $(x^1, \ldots, x^n)$, the volume form is expressed as:*

$$dV = \sqrt{\det(g_{ij})}\, dx^1 \wedge \cdots \wedge dx^n$$

*The corresponding measure for integration is often denoted as the **Riemannian density**:*

$$d\mu = \sqrt{\det g}\, d^n x$$

3.3 Length and Distance

Let $\gamma : [a, b] \to M$ be a smooth curve.

- **Length:** The length of γ is defined as:

$$L(\gamma) = \int_a^b |\dot{\gamma}(t)|_g \, dt = \int_a^b \sqrt{g(\dot{\gamma}(t), \dot{\gamma}(t))} \, dt$$

- **Invariance:** Length is invariant under reparametrization.
- **Regularity:** γ is *regular* if $\dot{\gamma}(t) \neq 0$ for all t (i.e., it is an immersion). γ is *admissible* if it is continuous and piecewise regular.
- **Arc-length Parameter:** $s(t) = \int_a^t |\dot{\gamma}(u)| du$. Under this reparametrization, $|\frac{d\gamma}{ds}| = 1$ (unit speed).

Proposition 3.3. *If $(M, g) \to (\tilde{M}, \tilde{g})$ is an isometry, then the induced map preserves lengths. i.e. $L(\gamma) = L(\Phi \circ \gamma)$.*

Distance Definition: For a connected Riemannian manifold (M, g), the distance $d(p, q)$ between $p, q \in M$ is defined as:

$$d(p, q) = \inf\{L(\gamma) \mid \gamma : [a, b] \to M \text{ is admissible}, \gamma(a) = p, \gamma(b) = q\}$$

This definition exists since the manifold is connected and the path is compact.

Theorem 3.4. *(M, d) is a metric space, and the metric topology coincides with the manifold topology.*

The key step is to show that in local coordinates, the Riemannian metric is equivalent to the Euclidean metric.

Lemma 3.5. *Let (M, g) be Riemannian manifold and $(U, \{x^1, ..., x^n\})$ be a local coordinate system. Set $\bar{g}$ be the Euclidean metric on U defined by $\bar{g} = \sum (dx^i)^2$. Then for any compact set $K \subset U$, and $x \in K$, $v \in T_x M$, we have*

$$|v|_{\bar{g}} \approx_{C_k} |v|_g.$$

Proof. Consider the continuous function $F : TU \to \mathbb{R}$ by $F(x, v)|v|_g$. Since K is compact, F attains a minimum and maximum on the set $\{(x, v) \in TU | x \in K, |v|_{\bar{g}} = 1\}$. □

3.4 Maximally Symmetric Spaces

Definitions

- **Homogeneous:** The isometry group $\mathrm{Isom}(M, g)$ acts transitively on M.
- **Isotropic:** The isometry group $\mathrm{Isom}(M, g)$ acts transitively on the unit tangent bundle $T^1 M$. Recall the action is given by $d\Phi_p(v)$ for $\Phi \in \mathrm{Isom}(M, g)$, $p \in M$, and $v \in T_p^1 M$.
- **Frame Isotropic:** $\mathrm{Isom}(M, g)$ acts transitively on the bundle of orthonormal frames $O(M)$. This implies M has constant sectional curvature.

3.5 Examples of Riemannian Manifolds with Maximal Symmetry

1. **Euclidean Space $\mathbb{R}^n$:** Standard metric $g = \sum (dx^i)^2$. $\mathrm{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$. $\dim = n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$.

2. **Sphere S^n:** Metric induced from $\mathbb{R}^{n+1}$. $\mathrm{Isom}(S^n) = O(n+1)$. $\dim = \frac{n(n+1)}{2}$.

3. **Hyperbolic Space $\mathbb{H}^n$:** Constant curvature $K = -1$.

   - **Hyperboloid Model:** In $\mathbb{R}^{n,1}$, surface $\tau^2 - \sum (\xi^i)^2 = 1, \tau > 0$.
   - **Poincaré Ball Model:** $B^n = \{x \in \mathbb{R}^n \mid |x| < 1\}$, $g = \frac{4\sum (dx^i)^2}{(1-|x|^2)^2}$.
   - **Upper Half-Plane Model:** $U^n = \{(x_1, \ldots, y) \in \mathbb{R}^n \mid y > 0\}$, $g = \frac{\sum (dx^i)^2 + dy^2}{y^2}$.

   $\mathrm{Isom}(\mathbb{H}^n) \cong O(n,1)$. $\dim = \frac{n(n+1)}{2}$.

4 Lecture 5

4.1 Three Models of Hyperbolic Space $\mathbb{H}^n$

Hyperbolic space can be represented by several isometric models. We focus on the Upper Half-Plane model ($\mathbb{U}^n$), the Poincaré Ball model ($\mathbb{B}^n$), and the Hyperboloid model.

Metrics and Transformations

- **Upper Half-Plane Model** $\mathbb{U}^n$: Defined on $\{(x_1, \ldots, y) \in \mathbb{R}^n : y > 0\}$. In dimension 2 ($z = x + iy$):

$$g_{\mathbb{U}} = \frac{dx^2 + dy^2}{y^2} = \frac{dz d\bar{z}}{(\operatorname{Im} z)^2}$$

- **Poincaré Ball Model** $\mathbb{B}^n$: Defined on $\{w \in \mathbb{R}^n : |w| < 1\}$. In dimension 2 ($w = u + iv$):

$$g_{\mathbb{B}} = 4\frac{du^2 + dv^2}{(1 - |w|^2)^2} = 4\frac{dw d\bar{w}}{(1 - |w|^2)^2}$$

Cayley Transform: The isometry $X : \mathbb{B}^2 \to \mathbb{U}^2$ is given by the Möbius transformation:

$$z = X(w) = -i\frac{w + i}{w - i}$$

Verification of Isometry. 1. Compute the differential dz:

$$dz = -i\frac{(w - i) - (w + i)}{(w - i)^2}dw = -i\frac{-2i}{(w - i)^2}dw = \frac{-2}{(w - i)^2}dw$$

Thus, $dz d\bar{z} = \frac{4}{|w-i|^4} dw d\bar{w}$.

2. Compute the imaginary part $\operatorname{Im} z$:

$$\operatorname{Im} z = \frac{1}{2i}(z - \bar{z}) = \frac{1}{2i}\left(-i\frac{w + i}{w - i} - i\frac{\bar{w} - i}{\bar{w} + i}\right) = -\frac{1}{2}\left(\frac{(w + i)(\bar{w} + i) + (\bar{w} - i)(w - i)}{|w - i|^2}\right)$$

$$\operatorname{Im} z = -\frac{1}{2}\left(\frac{(w\bar{w} + iw + i\bar{w} - 1) + (w\bar{w} - iw - i\bar{w} - 1)}{|w - i|^2}\right) = \frac{1 - |w|^2}{|w - i|^2}$$

3. Substitute into the metric $g_{\mathbb{U}}$:

$$\frac{dz d\bar{z}}{(\operatorname{Im} z)^2} = \frac{4/|w - i|^4}{(1 - |w|^2)^2/|w - i|^4} dw d\bar{w} = \frac{4 dw d\bar{w}}{(1 - |w|^2)^2} = g_{\mathbb{B}}$$

□

4.2 Isometry Groups

The groups of orientation-preserving isometries for these models are isomorphic:

- $\operatorname{Isom}^+(\mathbb{U}^2) \cong PSL(2, \mathbb{R}) = \{\begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R}, ad - bc = 1\}/\{\pm I\}$.
- $\operatorname{Isom}^+(\mathbb{B}^2) \cong PSU(1, 1) = \{\begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1\}/\{\pm I\}$.
- $\operatorname{Isom}^+(\mathbb{H}^2) \cong SO^+(2, 1)$.

Hence, $PSL(2, \mathbb{R}) \cong PSU(1, 1) \cong SO^+(2, 1)$.

4.3 Geometric Schwarz Lemma

Schwarz Lemma (Classical): If $f : \mathbb{B}^2 \to \mathbb{B}^2$ is holomorphic and $f(0) = 0$, then $|f'(0)| \leq 1$.

Theorem 4.1 (Geometric Schwarz Lemma)**.** *If $f : \mathbb{B}^2 \to \mathbb{B}^2$ is a holomorphic map, then for all $w_1, w_2 \in \mathbb{B}^2$:*

$$d_{hyp}(f(w_1), f(w_2)) \leq d_{hyp}(w_1, w_2)$$

where d_{hyp} is the distance induced by the hyperbolic metric.

Proof. **Step 1:** Show that for any $X \in T_p\mathbb{B}^2$, $|df_p(X)|_{\text{hyp}} \leq |X|_{\text{hyp}}$. Since $\text{Isom}^+(\mathbb{B}^2)$ acts transitively, we can assume $p = 0$ and $f(p) = 0$ by pre- and post-composing with isometries. At the origin, the hyperbolic metric is related to the Euclidean metric by $|X|_{\text{hyp}} = 2|X|_{\text{euc}}$. By the classical Schwarz Lemma, $|f'(0)| \leq 1$. Thus:

$$|df_0(X)|_{\text{hyp}} = 2|f'(0)X|_{\text{euc}} = |f'(0)| \cdot 2|X|_{\text{euc}} \leq |X|_{\text{hyp}}$$

Step 2: For any curve γ, $L(f \circ \gamma) \leq L(\gamma)$. Since the distance is the infimum of lengths of admissible curves, the inequality $d_{\text{hyp}}(f(w_1), f(w_2)) \leq d_{\text{hyp}}(w_1, w_2)$ follows. □

4.4 Connections on Vector Bundles

Directional Derivative on Functions

For $F \in C^\infty(M)$, $X \in \mathfrak{X}(M)$, $D_X F = XF$ satisfies: 1. $\mathbb{R}$-linearity in X and F. 2. C^∞-linearity in X: $D_{fX}F = fD_X F$. 3. Leibniz rule: $D_X(fF) = (Xf)F + f(XF)$.

The Connection ∇

Let $E \to M$ be a real vector bundle. A **connection** on E is a map $\nabla : \mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$, denoted $(X, \xi) \mapsto \nabla_X \xi$, satisfying:

1. $\nabla_X \xi$ is $\mathbb{R}$-linear in X and ξ.
2. $C^\infty(M)$**-linearity in** X: $\nabla_{fX}\xi = f\nabla_X \xi$.
3. **Leibniz Rule in** ξ: $\nabla_X(f\xi) = (Xf)\xi + f\nabla_X \xi$.

Proposition 4.2. *$\nabla_X \xi(p)$ depends only on the value of X at p and the values of ξ in an arbitrarily small neighborhood of p.*

Proof. In a coordinate neighborhood U with $X = \sum X^i \partial_i$, we have $\nabla_X \xi = \sum X^i \nabla_{\partial_i} \xi$. If $X(p) = 0$, then $X^i(p) = 0$, so $\nabla_X \xi(p) = 0$. This shows point-wise dependence on X. For ξ, if $\xi \equiv \tilde{\xi}$ on U, then $\nabla_X(\xi - \tilde{\xi}) = 0$ at p by the Leibniz rule. □

Curvature Tensor

Definition 4.3. The curvature $F(X, Y)\xi$ (often denoted $R(X, Y)\xi$) is defined as:

$$F(X, Y)\xi = \nabla_X \nabla_Y \xi - \nabla_Y \nabla_X \xi - \nabla_{[X,Y]}\xi$$

It can be shown that F is $C^\infty(M)$-linear in X, Y, and ξ, hence it is a tensor field.

5 Lecture 6

5.1 The Curvature Tensor

Definition 5.1. Let ∇ be a connection on a vector bundle $E \to M$. The **curvature operator** $F(X,Y) : \Gamma(E) \to \Gamma(E)$ is defined by:

$$F(X,Y)\xi = \nabla_X \nabla_Y \xi - \nabla_Y \nabla_X \xi - \nabla_{[X,Y]}\xi$$

for vector fields $X, Y \in \mathfrak{X}(M)$ and a section $\xi \in \Gamma(E)$.

Proposition 5.2. *F is $C^\infty(M)$-multilinear in X, Y, ξ.*

Proof for linearity in ξ. Let $f \in C^\infty(M)$. We want to show $F(X,Y)(f\xi) = fF(X,Y)\xi$.

$$\begin{aligned}
F(X,Y)(f\xi) &= \nabla_X \nabla_Y (f\xi) - \nabla_Y \nabla_X (f\xi) - \nabla_{[X,Y]}(f\xi) \\
&= \nabla_X \left((Yf)\xi + f\nabla_Y \xi\right) - \nabla_Y \left((Xf)\xi + f\nabla_X \xi\right) - \big(([X,Y]f)\xi + f\nabla_{[X,Y]}\xi\big) \\
&= (X(Yf)\xi + (Yf)\nabla_X \xi + (Xf)\nabla_Y \xi + f\nabla_X \nabla_Y \xi) \\
&\quad - (Y(Xf)\xi + (Xf)\nabla_Y \xi + (Yf)\nabla_X \xi + f\nabla_Y \nabla_X \xi) \\
&\quad - \big((X(Yf) - Y(Xf))\xi + f\nabla_{[X,Y]}\xi\big) \\
&= f\left(\nabla_X \nabla_Y \xi - \nabla_Y \nabla_X \xi - \nabla_{[X,Y]}\xi\right) = fF(X,Y)\xi
\end{aligned}$$

The terms involving derivatives of f cancel out completely. □

Corollary 5.3. *F is a section of the bundle $\Lambda^2 M \otimes End(E)$. If $E = TM$, we call ∇ a **linear connection**.*

5.2 Linear Connections in Local Coordinates

Let $\{E_i\}$ be a local frame for TM (typically coordinate vectors ∂_i). We define the **Christoffel symbols** Γ_{ij}^k by:

$$\nabla_{E_i} E_j = \Gamma_{ij}^k E_k$$

For vector fields $X = X^i E_i$ and $Y = Y^j E_j$, the covariant derivative is:

$$\nabla_X Y = \sum_k \left(X(Y^k) + \sum_{i,j} \Gamma_{ij}^k X^i Y^j \right) E_k$$

Notation: In coordinate basis ∂_i, we often denote:

$$\nabla_{\partial_i}(V^j \partial_j) = V^j_{;i} \partial_j, \quad \text{where } V^j_{;i} = \partial_i V^j + \Gamma_{ik}^j V^k$$

5.3 Induced Connections

Connection on the Dual Bundle T^*M

We wish to define a connection on T^*M such that the Leibniz rule holds for the pairing $\omega(Y)$:

$$X(\omega(Y)) = (\nabla_X \omega)(Y) + \omega(\nabla_X Y)$$

Definition 5.4. The induced connection on T^*M is given by:

$$(\nabla_X \omega)(Y) = X(\omega(Y)) - \omega(\nabla_X Y)$$

Proposition 5.5. *This defines a valid connection on T^*M.*

Verification of 1-form property. For $f \in C^\infty(M)$, we must show $(\nabla_X \omega)(fY) = f(\nabla_X \omega)(Y)$:

$$\begin{aligned}(\nabla_X \omega)(fY) &= X(\omega(fY)) - \omega(\nabla_X(fY)) \\ &= X(f\omega(Y)) - \omega((Xf)Y + f\nabla_X Y) \\ &= (Xf)\omega(Y) + fX(\omega(Y)) - (Xf)\omega(Y) - f\omega(\nabla_X Y) \\ &= f\left(X(\omega(Y)) - \omega(\nabla_X Y)\right) = f(\nabla_X \omega)(Y)\end{aligned}$$

Similar proofs hold for C^∞-linearity in X and the Leibniz rule in ω. □

Local Coordinate Form: For $\omega = \omega_j dx^j$, the covariant derivative is:

$$\nabla_{\partial_i}(\omega_j dx^j) = \omega_{j;i} dx^j, \quad \text{where } \omega_{j;i} = \partial_i \omega_j - \Gamma^k_{ij}\omega_k$$

Connection on Tensor Products

A connection can be induced on the tensor bundle $TM \otimes TM$:

$$\nabla_X(Y \otimes Z) = (\nabla_X Y) \otimes Z + Y \otimes (\nabla_X Z)$$

Covariant Derivative Along a Curve

Setup: Let $\gamma : I \to M$ be a smooth curve.

- Velocity vector field: $\dot{\gamma}(t) = d\gamma_t(\partial_t)$.
- Vector field along γ: A map $V : I \to TM$ such that $V(t) \in T_{\gamma(t)}M$. This is a section of the pullback bundle $\gamma^* TM$.
- **Extendability:** $V \in \mathfrak{X}(\gamma)$ is extendable if there exists a vector field $\tilde{V}$ in a neighborhood of γ such that $\tilde{V}_{\gamma(t)} = V(t)$.

Proposition: Given a connection ∇ on M, there is a unique operator $D_t : \mathfrak{X}(\gamma) \to \mathfrak{X}(\gamma)$ satisfying:

1. D_t is $\mathbb{R}$-linear.
2. Leibniz Rule: $D_t(fV) = \frac{df}{dt}V + fD_tV$ for $f \in C^\infty(I)$.
3. Consistency: If V extends to $\tilde{V}$, then $(D_tV)(t) = \nabla_{\dot{\gamma}(t)}\tilde{V}$.

Local Coordinate Form: If $V(t) = V^i(t)\partial_i$, then:

$$D_tV = \sum_k \left(\frac{dV^k}{dt} + \sum_{i,j} \Gamma^k_{ij} \frac{d\gamma^i}{dt} V^j \right) \partial_k$$

6 Lecture 7

6.1 Geodesics

Definition 6.1. A smooth curve $\gamma : I \to M$ is a **geodesic** with respect to a connection ∇ if its velocity vector field is parallel along the curve, i.e.,

$$D_t\dot{\gamma} = 0.$$

Local Coordinates: In a local coordinate system $\{x^i\}$, let $\gamma(t) = (\gamma^1(t), \ldots, \gamma^n(t))$. The velocity is $\dot{\gamma} = \frac{d\gamma^i}{dt}\partial_i$. The **geodesic equation** is a system of second-order non-linear ODEs:

$$\frac{d^2\gamma^k}{dt^2} + \sum_{i,j} \Gamma^k_{ij}(\gamma(t))\frac{d\gamma^i}{dt}\frac{d\gamma^j}{dt} = 0, \quad \text{for } k = 1, \ldots, n.$$

Existence and Uniqueness: By standard ODE theory (Picard-Lindelöf), given $p \in M$ and $v \in T_pM$, there exists a unique maximal geodesic $\gamma : I \to M$ such that $\gamma(0) = p$ and $\dot{\gamma}(0) = v$.

- **Example:** In $M = \mathbb{R}^2 \setminus \{0\}$, a geodesic starting at $(-1, 0)$ with velocity $(1, 0)$ is defined only for $t \in (-\infty, 1)$.

6.2 Parallel Transport

Definition 6.2. A vector field V along γ is **parallel** if $D_tV = 0$.

Parallel Transport Map: For a curve $\gamma : [t_0, t_1] \to M$, the map $P_{t_0,t_1} : T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M$ defined by $v_0 \mapsto V(t_1)$ (where V is parallel and $V(t_0) = v_0$) is a linear isomorphism.

Relationship to Covariant Derivative: The covariant derivative can be recovered from parallel transport:

$$(D_tV)(t_0) = \lim_{t\to t_0} \frac{P_{t,t_0}V(t) - V(t_0)}{t - t_0} = \left.\frac{d}{dt}\right|_{t=t_0} P_{t,t_0}V(t).$$

Proof Sketch: Use a parallel frame $\{E_i(t)\}$ such that $D_tE_i = 0$. Write $V(t) = v^i(t)E_i(t)$, then $P_{t,t_0}V(t) = v^i(t)E_i(t_0)$. Taking the derivative yields the component-wise derivative.

6.3 Connections on Submanifolds of $\mathbb{R}^n$

Let $M \subset \mathbb{R}^n$ be an embedded submanifold. Let $\Pi^\top : T_p\mathbb{R}^n \to T_pM$ denote the orthogonal projection onto the tangent space.

Induced Connection: Let $\bar{\nabla}$ be the standard Euclidean connection on $\mathbb{R}^n$. For vector fields X, Y on M, define:

$$\nabla_X Y = \Pi^\top(\bar{\nabla}_{\bar{X}}\bar{Y})$$

where $\bar{X}, \bar{Y}$ are local extensions of X, Y to $\mathbb{R}^n$.

Properties:

1. **Metric Compatibility:** $X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$.
2. **Symmetry (Torsion-free):** $\nabla_X Y - \nabla_Y X = [X, Y]$.

These properties confirm that the induced connection is the **Levi-Civita connection** for the induced Riemannian metric on M.

Proof of Connection Properties on Submanifolds

Let $M \subset \mathbb{R}^n$ be an embedded submanifold. Let $\bar{\nabla}$ denote the standard Euclidean connection on $\mathbb{R}^n$, and $\Pi^\top : T_p\mathbb{R}^n \to T_pM$ be the orthogonal projection onto the tangent space. The induced connection on M is defined as $\nabla_X Y = \Pi^\top(\bar{\nabla}_{\bar{X}}\bar{Y})$, where $\bar{X}, \bar{Y}$ are local extensions of X, Y.

Metric Compatibility

Property: $X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$.

Proof. The standard connection $\bar{\nabla}$ on $\mathbb{R}^n$ is compatible with the Euclidean metric, satisfying:

$$\bar{X}\langle \bar{Y}, \bar{Z}\rangle = \langle \bar{\nabla}_{\bar{X}}\bar{Y}, \bar{Z}\rangle + \langle \bar{Y}, \bar{\nabla}_{\bar{X}}\bar{Z}\rangle$$

When restricted to the manifold M, the term $\bar{X}\langle \bar{Y}, \bar{Z}\rangle$ becomes $X\langle Y, Z\rangle$. Since $Z \in T_pM$ is a tangent vector, the inner product only "sees" the tangential component of any vector it is paired with. Thus:

$$\langle \bar{\nabla}_{\bar{X}}\bar{Y}, Z\rangle = \langle \Pi^\top(\bar{\nabla}_{\bar{X}}\bar{Y}), Z\rangle = \langle \nabla_X Y, Z\rangle$$

Similarly, $\langle Y, \bar{\nabla}_{\bar{X}}\bar{Z}\rangle = \langle Y, \nabla_X Z\rangle$. Substituting these back into the Euclidean identity yields:

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$$

□

Symmetry (Torsion-free)

Property: $\nabla_X Y - \nabla_Y X = [X, Y]$.

Proof. The standard connection $\bar{\nabla}$ on $\mathbb{R}^n$ is torsion-free, meaning:

$$\bar{\nabla}_{\bar{X}}\bar{Y} - \bar{\nabla}_{\bar{Y}}\bar{X} = [\bar{X}, \bar{Y}]$$

Applying the orthogonal projection $\Pi^\top$ to both sides gives:

$$\Pi^\top(\bar{\nabla}_{\bar{X}}\bar{Y}) - \Pi^\top(\bar{\nabla}_{\bar{Y}}\bar{X}) = \Pi^\top([\bar{X}, \bar{Y}])$$

By the definition of the induced connection, the left-hand side is $\nabla_X Y - \nabla_Y X$. Furthermore, since M is a smooth submanifold, the Lie bracket of two tangent vector fields X and Y is itself tangent to M, implying $\Pi^\top([X, Y]) = [X, Y]$. Therefore:

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

□

7 Lecture 8

7.1 Holonomy Group

Definition 7.1. Let ∇ be a linear connection on M. Given a loop (curve) $\gamma : [t_0, t_1] \to M$ at $p = \gamma(t_0) = \gamma(t_1)$, the parallel transport $P_\gamma : T_pM \to T_pM$ is a linear automorphism. We note $H(\gamma) \subset Aut(T_pM)$ be the automorphisms induced by the loop γ. The **Holonomy group** at p is defined as:

$$\mathrm{Hol}_p(\nabla) = \{P_\gamma \in H(\gamma) \mid \gamma \text{ is a loop at } p\}.$$

Property: If p, p' are in the same connected component of M, then $\mathrm{Hol}_p(\nabla) \cong \mathrm{Hol}_{p'}(\nabla)$ via conjugation by parallel transport along a path connecting p and p'.

Example 7.2. In $\mathbb{R}^n$ with the standard connection, $\mathrm{Hol}_p(\nabla) = \{I\}$ for all p since parallel transport along any loop is the identity.

7.2 Metric Compatibility

Definition 7.3. A connection ∇ is **metric-compatible** if:

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle, \quad \forall X, Y, Z \in \mathfrak{X}(M).$$

Proposition 7.4. ***TFAE:***

1. *∇ is metric-compatible.*
2. *$\nabla g = 0$.*
3. *$\frac{d}{dt}\langle V, W\rangle = \langle D_t V, W\rangle + \langle V, D_t W\rangle$ for vector fields along any curve γ.*
4. *Parallel transport $P_{t_0,t_1} : T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M$ is an isometry.*
5. *$Hol_p(\nabla) \subset O(n)$ (after choosing an orthonormal basis).*

Proof. We follow the logic from the lecture notes to prove the equivalences.

$1 \iff 2$

Using the formula for the covariant derivative of a $(0, 2)$-tensor g:

$$X(g(Y, Z)) = (\nabla_X g)(Y, Z) + g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

Since $g(Y, Z) = \langle Y, Z\rangle$, $\nabla g = 0$ is clearly equivalent to the metric compatibility condition.

$1 \implies 3$

Define the operator $S(V, W) = \frac{d}{dt}\langle V, W\rangle - \langle D_t V, W\rangle - \langle V, D_t W\rangle$ for $V, W \in \mathfrak{X}(\gamma)$.

1. S is $C^\infty(I)$-linear. For $f \in C^\infty(I)$, the term f' from $\frac{d}{dt}(f\langle V, W\rangle)$ cancels with the f' from $\langle D_t(fV), W\rangle$.
2. Since S is linear, it suffices to check it on a local frame. For any t_0, extend $V(t_0), W(t_0)$ to vector fields Y, Z on M and let $X = \gamma'(t_0)$.
3. Then $S(V, W)(t_0) = X\langle Y, Z\rangle - \langle \nabla_X Y, Z\rangle - \langle Y, \nabla_X Z\rangle$, which is zero by condition 1. Thus $S \equiv 0$.

$3 \implies 4$

If V_1, V_2 are parallel, then $D_t V_1 = 0$ and $D_t V_2 = 0$. Substituting into condition 3: $\frac{d}{dt}\langle V_1, V_2\rangle = \langle 0, V_2\rangle + \langle V_1, 0\rangle = 0$. Thus the inner product is constant.

$4 \implies 3$

Let $\{E_i(t)\}$ be an orthonormal basis at $\gamma(t_0)$ extended by parallel transport along γ. By condition 4, $\{E_i(t)\}$ remains an orthonormal basis for all t (since $D_t E_i = 0$). Let $V(t) = V^i(t)E_i(t)$ and $W(t) = W^j(t)E_j(t)$. Then $\langle V, W\rangle = \sum V^i W^i$. Taking the derivative: $\frac{d}{dt}\langle V, W\rangle = \sum(\dot{V}^i W^i + V^i \dot{W}^i)$. Meanwhile, $D_t V = \dot{V}^i E_i$ implies $\langle D_t V, W\rangle = \sum \dot{V}^i W^i$. The identity holds.

$4 \iff 5$

By definition, $P_{t_0,t_1} V(t_0) = V(t_1)$ where V is a parallel vector field. Condition 4 states $\langle V(t_0), \tilde{V}(t_0)\rangle = \langle V(t_1), \tilde{V}(t_1)\rangle$, which means P_{t_0,t_1} preserves the inner product, hence it is an isometry. $\square$

7.3 Torsion and Levi-Civita Connection

Definition 7.5. The **Torsion tensor** T is defined by:

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

A connection is **torsion-free** if $T = 0$.

Theorem 7.6 (Fundamental Theorem of Riemannian Geometry)**.** *On any Riemannian manifold (M, g), there exists a unique connection ∇ that is both metric-compatible and torsion-free. This is the **Levi-Civita connection**.*

The one and only connection is given by:
Koszul Formula:

$$\begin{aligned} 2\langle \nabla_X Y, Z\rangle = {} & X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle \\ & + \langle [X, Y], Z\rangle - \langle [Y, Z], X\rangle + \langle [Z, X], Y\rangle. \end{aligned}$$

7.4 Invariant Metrics and Compact Groups

If the holonomy group $K = \mathrm{Hol}_p(\nabla)$ is a compact subgroup of $\mathrm{Aut}(T_p M)$, we can construct an invariant inner product by averaging over the Haar measure μ of K:

$$\widetilde{\langle v, w\rangle} = \int_K \langle kv, kw\rangle_0 \, d\mu(k).$$

This implies that a connection comes from some Riemannian metric if and only if its holonomy group is contained in a conjugate of $O(n)$.

8 Lecture 9

8.1 Holonomy Groups

Definition 8.1 (Restricted Holonomy Group)**.** The restricted holonomy group $\mathrm{Hol}_0(p)$ consists of parallel translations around contractible loops based at p.

fact:$\mathrm{Hol}_0(p)$ is the identity component of $\mathrm{Hol}(p)$. It is a connected, closed Lie subgroup of $SO(n)$.

Definition 8.2 (Special Holonomy)**.** Generally, $\mathrm{Hol}(p) = O(n)$. If $\mathrm{Hol}(p) \subsetneq O(n)$, the manifold is said to have *special holonomy* (e.g., Kähler, G2, etc.).

8.2 Levi-Civita Connection and Christoffel Symbols

Theorem 8.3 (Koszul Formula)**.** *The unique Levi-Civita connection is determined by:*

$$2\langle \nabla_X Y, Z\rangle = X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle + \langle [X, Y], Z\rangle - \langle [Y, Z], X\rangle + \langle [Z, X], Y\rangle$$

Remark 8.4 (In Local Coordinates). Let ∂_i be coordinate vectors. Then $\langle \nabla_{\partial_i}\partial_j, \partial_l\rangle = \frac{1}{2}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$. We define the Christoffel symbols of the first and second kind:

$$\Gamma_{ijl} = \langle \nabla_{\partial_i}\partial_j, \partial_l\rangle, \quad \nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\partial_k \implies \Gamma^k_{ij} = g^{kl}\Gamma_{ijl}$$

Remark 8.5 (Symmetry). Since $\nabla_{\partial_i}\partial_j - \nabla_{\partial_j}\partial_i = [\partial_i, \partial_j] = 0$, we have $\Gamma^k_{ij} = \Gamma^k_{ji}$.

8.3 Properties of Geodesics

Proposition 8.6. *If ∇ is the Levi-Civita connection and γ is a geodesic, then $|\dot\gamma|$ is constant.*

Proof. $\frac{d}{dt}|\dot\gamma|^2 = \frac{d}{dt}\langle\dot\gamma, \dot\gamma\rangle = 2\langle D_t\dot\gamma, \dot\gamma\rangle$. Since γ is a geodesic, $D_t\dot\gamma = 0$, hence the derivative is zero. □

8.4 Isometries and Geodesics

Proposition 8.7. *If $\phi : M \to \tilde M$ is an isometry, then:*

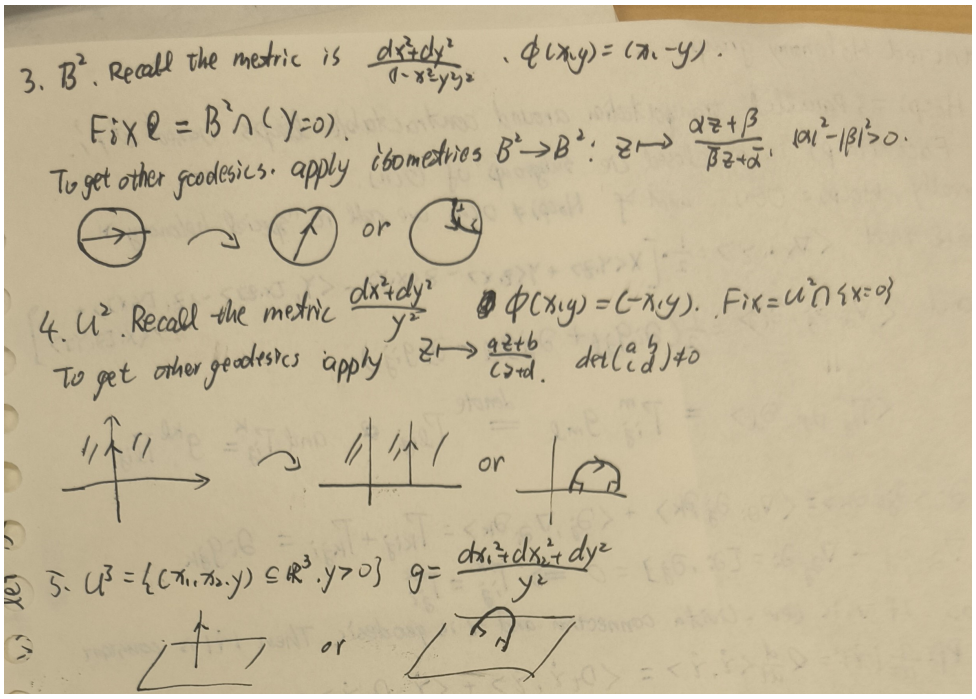
1. *$\phi_*(\nabla_X Y) = \tilde\nabla_{\phi_* X}\phi_* Y$. (Isometries preserve the Levi-Civita connection).*
2. *γ is a geodesic on $M \iff \phi\circ\gamma$ is a geodesic on $\tilde M$.*

Proof. **(1)** First, define $\phi_*\tilde\nabla$ by $(\phi_*\tilde\nabla)_X Y = \phi_*^{-1}(\tilde\nabla_{\phi_* X}\phi_* Y)$, and proof that it is a metric-compatible, torsion-free connection. Hence by unuqueness of Levi-Civita connection, $\phi_*\tilde\nabla = \nabla$.
(2) Trivial. □

Proposition 8.8 (Symmetry and Geodesics)**.** *If $\phi \in Isom(M)$ satisfies $\phi(p) = p$ and $d\phi_p(X) = X$, then the maximal geodesic γ with $\gamma(0) = p, \dot\gamma(0) = X$ is pointwise fixed by ϕ, i.e., $\phi(\gamma(t)) = \gamma(t)$.*

Example 8.9 (S^n)**.** Reflection through a 2-plane P spanned by p, X. The fixed point set $P\cap S^n$ is a great circle (geodesic).

Example 8.10 ($\mathbb{H}^n, \mathbb{B}^2, \mathbb{U}^2$)**.** Geodesics are obtained as fixed point sets of reflections (lines or circles orthogonal to the boundary).



8.5 The Exponential Map

Lemma 8.11 (Scaling)**.** *For a geodesic $\gamma_v(t)$, we have $\gamma_{cv}(t) = \gamma_v(ct)$.*

Definition 8.12 (Exponential Map)**.** Let $\mathcal{E} = \{v \in TM \mid \gamma_v(t) \text{ is defined for } t \in [0,1]\}$. The **exponential map** $\exp : \mathcal{E} \to M$ is defined by:

$$\exp(v) = \gamma_v(1), \quad \exp_p(v) = \exp|_{T_pM}(v)$$

Proposition 8.13 (Properties of $\exp_p$)**.**

1. *$\mathcal{E}$ is an open subset of TM containing the zero section.*
2. $\exp_p(tv) = \gamma_v(t)$.
3. $\exp_p$ *is smooth.*
4. $d(\exp_p)_0 : T_0(T_pM) \cong T_pM \to T_pM$ *is the identity map.*

Remark 8.14 (Motivation: Geodesic Flow). The geodesic equation $\ddot{x}^k + \Gamma^k_{ij}\dot{x}^i\dot{x}^j = 0$ can be written as a first-order system on TM:

$$\begin{cases} \frac{dx^i}{dt} = v^i \\ \frac{dv^k}{dt} = -\Gamma^k_{ij}v^iv^j \end{cases}$$

This defines a vector field G on TM whose flow gives the geodesics.

Proof of Properties of $\exp_p$. **$\mathcal{E}$ is an Open Subset of TM and $\exp$ is Smooth:**
Logic from Blackboard 2: Let $\Theta_t : \mathcal{D} \to TM$ be the flow generated by the vector field G. According to the theory of Ordinary Differential Equations (ODEs), the flow $\Theta_t(v)$ is defined on an open domain $\mathcal{D} \subset \mathbb{R} \times TM$.

We define the domain of the exponential map as:

$$\mathcal{E} = \{v \in TM \mid \Theta_t(v) \text{ is defined for } t \in [0,1]\}$$

If the flow starting from v exists up to time 1, then by the **fundamental theorem of ODEs** (continuous dependence on initial conditions), the flow exists up to time 1 (and "a bit farther") for all vectors in a sufficiently small neighborhood of v in TM. Hence, $\mathcal{E}$ is an **open subset** of TM.

Furthermore, since the geodesic vector field G is smooth, its flow Θ_t is also smooth. The exponential map is defined as the composition:

$$\exp = \pi \circ \Theta_1$$

where $\pi : TM \to M$ is the natural projection $(x, v) \mapsto x$. As a composition of smooth maps, $\exp : \mathcal{E} \to M$ is **smooth**.

$\exp_p(tv) = \gamma_v(t)$:
By the definition of the exponential map, $\exp_p(v) = \gamma_v(1)$. Recall the **scaling property** (homogeneity) of geodesics: for any constant $c > 0$, the curve $t \mapsto \gamma_v(ct)$ is a geodesic with initial velocity $c\dot{\gamma}_v(0) = cv$. By uniqueness of geodesics, we have:

$$\gamma_{cv}(t) = \gamma_v(ct)$$

Setting $c = t$ and evaluating at 1, we obtain:

$$\exp_p(tv) = \gamma_{tv}(1) = \gamma_v(t \cdot 1) = \gamma_v(t)$$

This shows that $\exp_p$ maps the line segment in T_pM through the origin to the corresponding geodesic on M.

The Differential $(d\exp_p)_0 = \mathbf{id}_{T_pM}$:
We wish to compute the derivative of $\exp_p : T_pM \to M$ at the origin $0 \in T_pM$. We identify $T_0(T_pM)$ with T_pM itself. For any $v \in T_pM$, consider the path $c(t) = tv$ in T_pM. Note that $c(0) = 0$ and $c'(0) = v$.

By the definition of the differential:

$$(d\exp_p)_0(v) = \left.\frac{d}{dt}\right|_{t=0} \exp_p(c(t)) = \left.\frac{d}{dt}\right|_{t=0} \exp_p(tv)$$

Using the property derived in the previous section ($\exp_p(tv) = \gamma_v(t)$):

$$(d\exp_p)_0(v) = \left.\frac{d}{dt}\right|_{t=0} \gamma_v(t) = \dot{\gamma}_v(0)$$

By the initial conditions of the geodesic γ_v, we have $\dot{\gamma}_v(0) = v$. Therefore:

$$(d\exp_p)_0(v) = v$$

which means $(d\exp_p)_0$ is the **identity map** on T_pM. $\square$

9 Lecture 10

9.1 Exponential Map and Normal Coordinates

Lemma 9.1. *For every $p \in M$, there exists a neighborhood V of $0 \in T_pM$ and a neighborhood $\mathcal{U}$ of p such that $\exp_p : V \to \mathcal{U}$ is a diffeomorphism.*

Definition 9.2 (Normal Neighborhood)**.** A neighborhood of p is called a **normal neighborhood** if it is the diffeomorphic image of a star-shaped neighborhood of $0 \in T_pM$ under $\exp_p$.

Definition 9.3 (Geodesic Ball)**.** If $\varepsilon > 0$ is small enough such that $\exp_p : B_\varepsilon(0) \to B_\varepsilon(p)$ is a diffeomorphism, $B_\varepsilon(p)$ is called a **geodesic ball**. Its boundary is the **geodesic sphere**.

Riemannian Normal Coordinates (RNC)

Let $\{E_i\}$ be an orthonormal basis for T_pM. The coordinate map is $(x^i) \mapsto \exp_p(\sum x^i E_i)$.

Proposition 9.4 (Properties of RNC)**.** *1. p has coordinates $(0, \dots, 0)$.*

2. Geodesics through p are radial lines: $x^i(t) = tv^i$.

3. At the origin: $g_{ij}(0) = \delta_{ij}$.

4. At the origin: $\Gamma^k_{ij}(0) = 0$.

Proof. 1 and 2 are automatic.

3. Consider the pull back metric $\exp_p^* g$, we aim to show that $\exp_p^* g(0) = \delta_{ij}$. For $V = V^i \frac{\partial}{\partial x^i} \in T_0 T_pM$. Put $\gamma(t) = tV$, $\exp_p(\gamma(t)) = \gamma_V(t)$. Then $(d\exp_p)_0(V) = \dot\gamma_V(0) = V$, i.e.

$$\exp_p^* g_0(V, V) = \|V\|^2 \Longrightarrow \exp_p^* g_0 = \delta_{ij}.$$

4. Let $\gamma(t) = tv$ be a geodesic. The geodesic equation is $\ddot x^i + \Gamma^i_{jk}\dot x^j \dot x^k = 0$. Substituting $x^i(t) = tv^i$, we have $0 + \Gamma^i_{jk}(0)v^j v^k = 0$. Since this holds for all $v \in T_pM$ and Γ^i_{jk} is symmetric, $\Gamma^i_{jk}(0) = 0$. □

9.2 Uniformly Normal Neighborhoods

Definition 9.5 (Uniformly Normal Neighborhood)**.** An open set $W \subset M$ is **uniformly normal** if there exists $\delta > 0$ such that for any $w \in W$, $W \subset \exp_w(B_\delta(0))$, and $\exp_w$ is a diffeomorphism on $B_\delta(0)$.

Proposition 9.6. *Every $p \in M$ has a uniformly normal neighborhood.*

Proof. Define the map $F : \mathcal{E} \to M \times M$ by $F(q, v) = (q, \exp_q v)$, where $\mathcal{E} \subset TM$ is the open domain where the exponential map is defined. We want to show that $dF_{(p,0)}$ is an isomorphism.

Recall that the tangent space $T_{(p,0)}TM$ fits into a short exact sequence involving the vertical space (tangent to the fiber T_pM) and the horizontal space (projecting to T_pM via $d\pi$). The tangent space to the product manifold is $T_{(p,p)}(M \times M) \cong T_pM \oplus T_pM$.

We can construct the following diagram:

$$\begin{tikzcd}
0 \arrow[r] & T_pM \arrow[r, "i_*"] \arrow[d, "\mathrm{id}"] & T_{(p,0)}TM \arrow[r, "d\pi"] \arrow[d, "dF_{(p,0)}"] & T_pM \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & T_pM \arrow[r, "0\oplus\mathrm{id}"] & T_{(p,p)}(M \times M) \arrow[r, "d\mathrm{pr}_1"] & T_pM \arrow[r] & 0
\end{tikzcd}$$

Let us verify that this diagram commutes:

- **Left Square:** For any $v \in T_pM$, the vertical lift $i_*(v)$ is the tangent vector to the curve $c(t) = (p, tv)$ in TM at $t = 0$. Applying the differential dF, we have:

$$dF_{(p,0)}(i_*(v)) = \frac{d}{dt}\bigg|_{t=0} F(p, tv) = \frac{d}{dt}\bigg|_{t=0} (p, \exp_p(tv)) = (0, v)$$

  which exactly matches the map $0 \oplus \mathrm{id}$ in the bottom row. Hence, the left square commutes.

- **Right Square:** Notice that for any $(q, v) \in TM$, projecting the first component gives $\mathrm{pr}_1(F(q, v)) = \mathrm{pr}_1(q, \exp_q v) = q = \pi(q, v)$. Taking the differential of both sides at $(p, 0)$ yields $d\mathrm{pr}_1 \circ dF_{(p,0)} = d\pi$, meaning the right square commutes.

Since the rows are exact and the two outer vertical maps are isomorphisms (they are identity maps), the **Five Lemma** implies that the middle map $dF_{(p,0)} : T_{(p,0)}TM \to T_{(p,p)}(M \times M)$ is a linear isomorphism.

By the **Inverse Function Theorem**, F is a local diffeomorphism at $(p, 0)$. Thus, there exists a neighborhood $V \subset TM$ of $(p, 0)$ that is mapped diffeomorphically onto a neighborhood of (p, p) in $M \times M$.

We can choose a sufficiently small open neighborhood $W \subset M$ of p and a radius $\delta > 0$ such that $W \times W \subset F(V \cap \{|v| < \delta\})$. Then, for any pair $(w, w') \in W \times W$, there exists a unique tangent vector $v \in T_wM$ with $|v| < \delta$ such that $F(w, v) = (w, w')$. By definition of F, this means $\exp_w(v) = w'$.

Furthermore, because F is a diffeomorphism on this restricted domain, for each fixed $w \in W$, the map $\exp_w$ acts as a diffeomorphism from the δ-ball $B_\delta(0) \subset T_wM$ onto its image, which completely contains W. Therefore, W is uniformly normal. □

9.3 Geodesic as length-minimizing curves

Remark 9.7 (Recall). Regular curve $\dot{\gamma} \neq 0$, admissible curve: piecewise regular curve.

Definition 9.8. Given intervals $I, J \in \mathbb{R}$. A 1-parameter family of curves is a continuous map $\Gamma : J \times I \to M$. We write $\Gamma_s(t) = \Gamma(s, t)$ be a curve for each s.

Definition 9.9. A vector field along Γ is a continuous map $V : J \times I \to TM$ s.t. $\pi \circ V = \Gamma$.

Definition 9.10. A 1-parameter family of curves is admissible if:

1. J is open and $I = [a, b]$.
2. $\exists$ partition $a = a_0 < a_1 < \cdots < a_k = b$, so Γ is smooth on $J \times [a_{i-1}, a_i]$.
3. $\Gamma_s(t)$ is admissible for each s.

Remark 9.11. $\partial_t\Gamma$ may not be a vector field along Γ, but $\partial_s\Gamma$ is.

Definition 9.12. Given $\gamma : [a, b] \to M$, a variation of γ is a 1-parameter family of curves (admissible), and $\Gamma(0, t) = \gamma(t)$.

Definition 9.13. If Γ is a variation of γ, then the variation field of γ is the piecewise smooth vector field

$$V(t) = \frac{\partial \Gamma}{\partial s}(0, t) \quad \text{along } \gamma.$$

Definition 9.14. A vector field along γ is proper if $V(a) = V(b) = 0$.

Remark 9.15. If Γ is a proper variation of γ, then the variation field is a proper vector field along γ.

Lemma 9.16. *If γ is admissible, and V is a vector field along γ, then $\exists$ variation Γ of γ, so V is its variation field.*
If V is proper then we can make Γ be proper.

Lemma 9.17. *On rectangles $J \times [a_{i-1}, a_i]$,*

$$D_s \partial_t \Gamma = D_t \partial_s \Gamma.$$

Proof. Say Γ lives in a coordinate neighborhood. $\Gamma(s,t) = (x^1(s,t), \ldots, x^n(s,t))$.

$$\partial_t \Gamma = \sum \frac{\partial x^i}{\partial t} \partial_i, \quad \partial_s \Gamma = \sum \frac{\partial x^i}{\partial s} \partial_i$$

$$D_t V = \sum \left(\frac{dv^i}{dt} + \Gamma^i_{jk} \frac{dx^j}{dt} v^k \right) \partial_i$$

Hence,

$$D_s \partial_t \Gamma = \sum \left(\frac{\partial^2 x^i}{\partial s \partial t} + \Gamma^i_{jk} \frac{\partial x^j}{\partial s} \frac{\partial x^k}{\partial t} \right) \partial_i$$

which is symmetric in s, t. $\square$

10 Lecture 11

10.1 Isometries and Geodesics

Proposition 10.1. *M, M' Riemannian manifold, s.t.*

1. *$p \in M$.*
2. *$\sigma_1, \sigma_2 : M \to M'$ isometries.*
3. *$\sigma_1(p) = \sigma_2(p)$.*
4. *$(d\sigma_1)_p = (d\sigma_2)_p$.*
5. *Every point $q \in M$ can be connected to p by a geodesic.*

then $\sigma_1 = \sigma_2$.

Proof. Put $\alpha = \sigma_1^{-1} \circ \sigma_2 : M \to M$ isometry. $\alpha(p) = p$, $d\alpha_p = \mathrm{id} : T_pM \to T_pM$.
Pick $q \in M$. Let $\gamma : [0, L] \to M$ be the geodesic connecting p and q.
$\alpha \circ \gamma$ is another geodesic with

$$(\alpha \circ \gamma)(0) = p, \quad (\alpha \circ \gamma)'(0) = \gamma'(0)$$

so $\alpha \circ \gamma = \gamma$. Hence $\alpha(q) = q$.
i.e., α is id. □

$\to$ We can use this prop to proof that $\mathrm{Isom}(S^n) = O(n+1)$.

Pf: Let $\sigma \in \mathrm{Isom}(S^n)$. $p \in S^n$, $\{E_i\}$ is ortho basis of T_pS^n.
Get $\sigma(p) \in S^n$. $\{(d\sigma)_p(E_i)\}$ ortho basis for $T_{\sigma(p)}S^n$.
$\exists A \in O(n+1)$ so

$$A(\vec{p}, \vec{E}_1, \dots, \vec{E}_n) = (\vec{\sigma}(p), \overrightarrow{d\sigma_p E_1}, \dots),$$

which uniquely determines the isom A.

10.2 The First Variation Formula

Theorem 10.2 (First Variation Formula)**.** *Let $\Gamma(s,t)$ be a variation of a unit-speed curve $\gamma(t)$. Let $L(s) = L(\Gamma_s)$ be the length of the curve $t \mapsto \Gamma(s,t)$. Then the first variation of arc length is given by:*

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \langle V(b), \dot{\gamma}(b)\rangle - \langle V(a), \dot{\gamma}(a)\rangle - \int_a^b \langle V(t), D_t\dot{\gamma}(t)\rangle \, dt$$

where D_t is the covariant derivative along γ.

Proof. Let $T = \partial_t\Gamma$ and $S = \partial_s\Gamma$. Note that at $s = 0$, $T = \dot{\gamma}$ and $S = V$. Furthermore, since γ is unit-speed, $|\dot{\gamma}| = 1$. The length function is:

$$L(s) = \int_a^b \langle T, T\rangle^{1/2} \, dt$$

Taking the derivative with respect to s:

$$\frac{d}{ds}L(s) = \int_a^b \frac{\partial}{\partial s}\langle T, T\rangle^{1/2}\, dt = \int_a^b \frac{1}{2}\langle T, T\rangle^{-1/2} \cdot 2\langle D_s T, T\rangle\, dt$$

Here we use the fact that the connection is metric-compatible, so $\partial_s\langle T, T\rangle = 2\langle D_s T, T\rangle$. Evaluating at $s = 0$, since $\langle T, T\rangle^{1/2} = |\dot{\gamma}| = 1$, we have:

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \int_a^b \langle D_s T, T\rangle\, dt$$

By the **Symmetry Lemma**, $D_s T = D_s(\partial_t \Gamma) = D_t(\partial_s \Gamma) = D_t S$. Substituting this in:

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \int_a^b \langle D_t S, T\rangle\, dt$$

Using the product rule for the covariant derivative: $\frac{\partial}{\partial t}\langle S, T\rangle = \langle D_t S, T\rangle + \langle S, D_t T\rangle$, we can integrate by parts:

$$\begin{aligned}\int_a^b \langle D_t S, T\rangle\, dt &= \int_a^b \left(\frac{\partial}{\partial t}\langle S, T\rangle - \langle S, D_t T\rangle\right) dt \\ &= \Big[\langle S, T\rangle\Big]_a^b - \int_a^b \langle S, D_t T\rangle\, dt\end{aligned}$$

Evaluating at $s = 0$ where $S = V$, $T = \dot{\gamma}$, and $D_t T = D_t\dot{\gamma}$, we obtain the desired formula:

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \langle V(b), \dot{\gamma}(b)\rangle - \langle V(a), \dot{\gamma}(a)\rangle - \int_a^b \langle V(t), D_t\dot{\gamma}\rangle\, dt$$

□

In the general case where γ is admissible, and $(a_1, ..., a_{k-1})$ are the break points, then its first variation formula is:

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \sum_{i=1}^{k-1}\langle V(a_i), \dot{\gamma}(a_i^+) - \dot{\gamma}(a_i^-)\rangle + \langle V(b), \dot{\gamma}(b)\rangle - \langle V(a), \dot{\gamma}(a)\rangle - \int_a^b \langle V(t), D_t\dot{\gamma}\rangle dt$$

We sometimes write $\dot{\gamma}(a_i^+) - \dot{\gamma}(a_i^-)$ as $\Delta_i\dot{\gamma}$.

Corollary 10.3. *Any length-minimizing curve between two points (parameterized by arc length) is a smooth geodesic.*

Proof. Let γ be a length-minimizing curve p and q. Fix t and let $\phi \in C^\infty([a, b])$ so that $\phi > 0$ on the smooth segment (a_{i-1}, a_i) for each i and $\phi = 0$ elsewhere.

Take $V(t) = \phi(t)D_t\dot{\gamma}(t)$ be a proper variation field, and $\Gamma(s, t) = \exp_{\gamma(t)}(sV(t))$ be a proper variation of γ on (a_{i-1}, a_i). Then the first variation formula gives:

$$0 = \left.\frac{d}{ds}\right|_{s=0} L(\Gamma_s) = -\int_{a_{i-1}}^{a_i} \phi(t)\langle D_t\dot{\gamma}(t), D_t\dot{\gamma}(t)\rangle\, dt \leq 0$$

Hence, $D_t\dot{\gamma} = 0$, i.t. γ is a geodesic on each smooth segment.

Next, consider the whole curve γ. At each break point a_i, we construct a proper variation field V so that $V(a_i) = \Delta_i\dot{\gamma}$ and equals 0 outside a small neighborhood of a_i. Then the first variation formula gives:

$$0 = \left.\frac{d}{ds}\right|_{s=0} L(\Gamma_s) = -\langle V(a_i), \Delta_i\dot{\gamma}\rangle = -|\Delta_i\dot{\gamma}|^2 \leq 0$$

Hence, $\Delta_i\dot{\gamma} = 0$, i.e. the geodesic segments on either side of a_i match up smoothly. Therefore, γ is a smooth geodesic.

□

10.3 Geodesic Polar Coordinates and Gauss Lemma

Definition 10.4 (Geodesic Polar Coordinates)**.** Let $p \in M$. For a point in a normal neighborhood of p, we can assign polar coordinates (r, θ) where $r > 0$ and $\theta \in S^{n-1} \subset T_pM$, via the mapping:

$$(r, \theta) \mapsto \exp_p(r\theta)$$

The coordinate fields are ∂_r (radial direction) and ∂_{θ^i} (tangent to the geodesic sphere $S_r(p)$).

Proposition 10.5. *On a normal basis, $|\partial_r| = 1$. Where, $\partial_r = \frac{x^i}{r(x)} \frac{\partial}{\partial x^i}$*

Lemma 10.6 (Gauss Lemma)**.** *In a normal neighborhood of p, the radial geodesics are orthogonal to the geodesic spheres centered at p. That is, $\langle \partial_r, \partial_{\theta^i} \rangle = 0$.*

Proof. Let $\sigma : (-\epsilon, \epsilon) \to S^{n-1} \subset T_pM$ be a curve on the unit sphere in the tangent space. Define a variation by radial geodesics:

$$\Gamma(s, t) = \exp_p(t\sigma(s))$$

Here, $T = \partial_t \Gamma = \partial_r$ is the tangent vector to the radial geodesics, and $S = \partial_s \Gamma = \partial_\theta$ represents a vector field tangent to the geodesic spheres. Notice that for each fixed s, $t \mapsto \exp_p(t\sigma(s))$ is a geodesic with constant speed $|\sigma(s)| = 1$. Therefore:

$$D_t T = D_t \partial_t \Gamma = 0 \quad \text{and} \quad \langle T, T \rangle = |\partial_t \Gamma|^2 = 1$$

We want to show that $\langle S, T \rangle = \langle \partial_s \Gamma, \partial_t \Gamma \rangle$ is independent of t. Taking the derivative with respect to t:

$$\frac{\partial}{\partial t} \langle \partial_s \Gamma, \partial_t \Gamma \rangle = \langle D_t \partial_s \Gamma, \partial_t \Gamma \rangle + \langle \partial_s \Gamma, D_t \partial_t \Gamma \rangle$$

By the Symmetry Lemma ($D_t \partial_s \Gamma = D_s \partial_t \Gamma$) and the fact that $D_t \partial_t \Gamma = 0$:

$$\frac{\partial}{\partial t} \langle S, T \rangle = \langle D_s \partial_t \Gamma, \partial_t \Gamma \rangle + 0 = \frac{1}{2} \frac{\partial}{\partial s} \langle \partial_t \Gamma, \partial_t \Gamma \rangle = \frac{1}{2} \frac{\partial}{\partial s} |\sigma(s)|^2 = \frac{1}{2} \frac{\partial}{\partial s}(1) = 0$$

This shows $\langle \partial_s \Gamma, \partial_t \Gamma \rangle$ is constant along the radial geodesic. To evaluate this constant, we look at $t = 0$. At $t = 0$, $\Gamma(s, 0) = \exp_p(0) = p$ for all s. Thus, the derivative with respect to s at $t = 0$ is the zero vector: $\partial_s \Gamma(s, 0) = 0$. Therefore, for all t:

$$\langle \partial_s \Gamma, \partial_t \Gamma \rangle|_t = \langle \partial_s \Gamma, \partial_t \Gamma \rangle|_{t=0} = \langle 0, \partial_t \Gamma \rangle = 0$$

Conclusion: $\langle \partial_r, \partial_\theta \rangle = 0$. □

Remark 10.7. In geodesic polar coordinates, the metric tensor takes the form:

$$g = dr^2 + \sum_{i,j=1}^{n-1} h_{ij}(r, \theta) d\theta^i d\theta^j$$

There are no $dr d\theta^i$ cross-terms because $\langle \partial_r, \partial_{\theta^i} \rangle = 0$, and the coefficient of dr^2 is 1 because radial geodesics have unit speed.

10.4 Geodesics are Locally Length-Minimizing

Proposition 10.8. *If $p \in M$ and q is a point in a geodesic ball around p, then the radial geodesic from p to q is the unique shortest curve from p to q.*

Proof. Let B be a normal neighborhood (geodesic ball) centered at p. The radial geodesic $\gamma_{p,q}$ connects p to q. Let R be the length of this radial geodesic (so $q = \exp_p(R\theta_0)$). Let $\sigma : [a, b] \to M$ be any other parameterized curve connecting p to q. Assume for now that σ lies entirely within B. In geodesic polar coordinates, we can write $\sigma(t)$ as:

$$\sigma(t) = \exp_p(r(t)\theta(t))$$

The velocity vector of σ is:

$$\dot{\sigma}(t) = \dot{r}(t)\partial_r + X(t)$$

where $X(t) = \sum \dot{\theta}^i(t)\partial_{\theta^i}$ is a vector tangent to the geodesic sphere $S_r(p)$. By the Gauss Lemma, $\partial_r \perp X(t)$. Thus, using the Pythagorean theorem for the inner product:

$$|\dot{\sigma}(t)|^2 = |\dot{r}(t)\partial_r|^2 + |X(t)|^2 = \dot{r}(t)^2 + |X(t)|^2 \geq \dot{r}(t)^2$$

Taking the square root, we have $|\dot{\sigma}(t)| \geq \dot{r}(t)$. The length of σ is:

$$L(\sigma) = \int_a^b |\dot{\sigma}(t)|\, dt \geq \int_a^b \dot{r}(t)\, dt = r(b) - r(a) = R - 0 = R$$

Hence, $L(\sigma) \geq R = L(\gamma_{p,q})$, proving the radial geodesic is minimizing.

Uniqueness: For equality to hold, $L(\sigma) = R$, we must have $|\dot{\sigma}(t)| = \dot{r}(t)$ almost everywhere. This implies two conditions:

1. $|X(t)| \equiv 0$, meaning $\dot{\theta}(t) = 0$, so $\theta(t)$ is constant. The curve must be a radial ray.
2. $\dot{r}(t) \geq 0$, meaning the curve never backtracks toward p.

Thus, the only curve achieving length R is the radial geodesic itself (up to monotonic reparameterization). □

Corollary 10.9. *1. A geodesic ball around p coincides with a metric ball around p (using the Riemannian distance function).*

2. Geodesics are locally length-minimizing: every point has a neighborhood such that any two points in the neighborhood are joined by a unique length-minimizing geodesic.

11 Lecture 12

11.1 Geodesics as Locally Minimizing Curves

Proposition 11.1. *A geodesic $\gamma : I \to M$ is locally minimizing. i.e., there exists a neighborhood U of t_0 such that for any $t_1, t_2 \in U$, the restriction $\gamma|_{[t_1,t_2]}$ is the unique shortest curve connecting $\gamma(t_1)$ and $\gamma(t_2)$.*

Proof. Take W, a uniformly normal neighborhood of $\gamma(t_0)$ with respect to parameter δ. By definition, for every $p \in W$, there exists a normal δ-ball around p, and W is completely contained within that normal ball.

Let U be the connected component of $\gamma^{-1}(W)$ containing t_0. For any $t_1, t_2 \in U$, we have $\gamma(t_1), \gamma(t_2) \in W$.

Because W is uniformly normal, we can center a normal δ-ball at $\gamma(t_1)$, and we are guaranteed that $\gamma(t_2)$ lies inside this normal ball. Within a normal geodesic ball, any point can be connected to the center by a unique radial geodesic, which is strictly shorter than any other curve connecting the two points. Since the original curve $\gamma|_{[t_1,t_2]}$ is a geodesic connecting $\gamma(t_1)$ and $\gamma(t_2)$ and lies within this normal ball, it must coincide with this minimizing radial geodesic (up to parameterization). Hence, $\gamma|_{[t_1,t_2]}$ minimizes the distance. □

11.2 Geodesic Completeness and the Hopf-Rinow Theorem

Definition 11.2. A Riemannian manifold M is **geodesically complete** if every maximal geodesic is defined for all $t \in \mathbb{R}$.

Example 11.3 (Counterexample)**.** $\mathbb{R}^2 \backslash \{(0,0)\}$ is not geodesically complete. A geodesic heading directly towards the removed origin cannot be extended for all $t \in \mathbb{R}$.

Theorem 11.4 (Hopf-Rinow Theorem - Part I)**.** *Let M be a connected Riemannian manifold. M is metrically complete (as a metric space) if and only if M is geodesically complete.*

Proof. $\Leftarrow$: Suppose (M, g) is metric-complete but not geodesic-complete. Then there exists a unit-speed maximal geodesic $\gamma : [a, b) \to M$ with $b < \infty$.

Since γ is unit-speed, the distance between any two points on the curve is bounded by the arc length:

$$d(\gamma(t), \gamma(t')) \leq |t - t'|$$

Take a sequence $t_j \to b$. Since $\{t_j\}$ is a Cauchy sequence in $\mathbb{R}$, the inequality above implies that $\{\gamma(t_j)\}$ is a Cauchy sequence in M.

Because M is assumed to be metrically complete, this Cauchy sequence must converge to some point $q \in M$, i.e., $\gamma(t_j) \to q$.

Now, take a uniformly normal neighborhood W of q with respect to some radius $\delta > 0$. For sufficiently large j, we have $\gamma(t_j) \in W$. By the property of the exponential map at $\gamma(t_j)$, we can extend the geodesic γ forward from $\gamma(t_j)$ by at least a distance of δ. However, since $t_j \to b$, we can choose j such that $b - t_j < \delta/2$. The extension will then go up to $t_j + \delta > b$, which contradicts the assumption that $[a, b)$ is the maximal domain of γ.

Hence, no such finite b can exist, and M is geodesically complete.

To proof the contrary, we will use the cliam below.

Claim: If $\exp_p$ is defined on all of T_pM for some $p \in M$, then for all $q \in M$, there exists a minimal geodesic connecting p to q.

Suppose $\{q_i\}$ is a Cauchy sequence in M. Assuming the claim holds, we can write $q_i = \exp_p(V_i)$ for some $V_i \in T_pM$, where the geodesic $\gamma_i(t) = \exp_p(tV_i)$ minimizes the distance, so $|V_i|_g = d(p, q_i)$.

Since $\{q_i\}$ is a Cauchy sequence, it is bounded in M. Therefore, the distances $d(p, q_i)$ are bounded, meaning there exists $D > 0$ such that $|V_i| \leq D$ for all i.

Because the sequence $\{V_i\}$ lies in a closed bounded set in the finite-dimensional vector space T_pM, by the Bolzano-Weierstrass theorem, there exists a convergent subsequence $V_{i(k)} \to W \in T_pM$.

Since the exponential map $\exp_p : T_pM \to M$ is continuous, we have:

$$\exp_p(V_{i(k)}) \to \exp_p(W) \quad \text{in the manifold topology.}$$

Because the original sequence $\{q_i\}$ is Cauchy and a subsequence converges to $\exp_p(W)$, the entire sequence must converge to the same limit. Hence, $\lim_{i\to\infty} q_i = \exp_p(W)$ in the metric topology. This proves M is a complete metric space. $\square$

Pf of the claim.
Obs: If $\gamma : [0,T] \to M$ is a minimizing geodesic from p to q. Then for all $b \in [0,T]$, $d(p,\gamma(b)) + d(\gamma(b),q) = d(p,q)$.
We call $\gamma : [0,b] \to M$ a <u>geodesic segment</u> with $\gamma(0) = p$ and we say γ <u>aims towards q</u> if γ is maximal between p and $\gamma(b)$, and $d(p,\gamma(b)) + d(\gamma(b),q) = d(p,q)$.
So its enough to show there is such a geodesic $\gamma : p \to \gamma(b)$ that towards for q with

$$L(\gamma) = d(p,q). \quad (= L(\gamma) + d(\gamma(b),q) \implies \gamma(b) = q)$$

To show this, take $\varepsilon > 0$. so $\overline{B_\varepsilon(p)}$ is a closed normal ball.
If $q \in \overline{B_\varepsilon(p)}$. then $\exists$ minimizing geodesic from p to q. Done.
If $q \notin \overline{B_\varepsilon(p)}$. Choose $x \in \partial B_\varepsilon(p)$ which minimizes the distance to q. We have unit-speed geodesic $\gamma : p \to x$. and we extend it to all $t \in \mathbb{R}$.
Since $d(p,q) = d(p,x) + d(x,q)$, $\gamma|_{[0,\varepsilon]}$ aims towards q.
Put $T = d(p,q)$, $\mathcal{A} = \{b \in [0,T] : \gamma|_{[0,b]} \text{ aims towards } q\}$.
$\mathcal{A}$ is a closed subset of $[0,T]$. Let $A = \max \mathcal{A}$.
If $A = T$. done. otherwise put $y = \gamma(A) \neq q$.
Take normal δ-Ball $\overline{B_\delta(y)}$ so $q \notin \overline{B_\delta(y)}$. Take a geodesic τ from y to $z \in \partial B_\delta(y)$. so τ aims towards q.

$$\begin{aligned} d(p,z) &\geq d(p,q) - d(z,q) && \text{So } \gamma|_{[0,A]} \cup \tau \text{ is minimizing curve } p \to z. \\ &= d(p,y) + d(y,z). && \text{hence its geodesic, aims towards } q. \end{aligned}$$

Corollaries of Hopf-Rinow:

1. If $\exists p \in M$ such that $\exp_p$ is defined on all of T_pM, then M is complete (metrically and geodesically).
2. If M is complete, then any two points can be connected by a minimizing geodesic.
3. M is Compact $\implies$ M is Geodesically complete. (Since compact metric spaces are complete).
4. If M is complete, $\exp_p : T_pM \to M$ is surjective for all $p \in M$.

11.3 Covering Maps and Local Isometries

Theorem 11.5. *Suppose $\tilde{M}, M$ are connected Riemannian manifolds, $\tilde{M}$ is complete, and $\pi : \tilde{M} \to M$ is a local isometry. Then M is complete, and π is a covering map.*

Recall: A map $\pi : \tilde{M} \to M$ is a covering map if it is surjective and for every $p \in M$, there exists a neighborhood U of p such that $\pi^{-1}(U) = \coprod_\alpha V_\alpha$ (disjoint union of open sets), and the restriction $\pi|_{V_\alpha} : V_\alpha \to U$ is a diffeomorphism.

Example 11.6 (Counterexample)**.** Let $\tilde{M} = (0, 10) \subset \mathbb{R}$, $M = S^1$, and $\pi(x) = e^{ix}$. Here, π is a local isometry but **not** a covering map. This is because $\tilde{M}$ is not complete.

Lemma 11.7 (Geodesic Lifting Lemma)**.** *If $p \in M$, $\tilde{p} \in \pi^{-1}(p)$, and $\gamma : I \to M$ is a geodesic, then there exists a unique geodesic $\tilde{\gamma}$ starting at $\tilde{p}$ such that $\pi \circ \tilde{\gamma} = \gamma$.*

Proof. Let $V = \gamma'(0) \in T_pM$. Since π is a local isometry, the differential $d\pi_{\tilde{p}} : T_{\tilde{p}}\tilde{M} \to T_pM$ is an isometric isomorphism. We can put $\tilde{V} = (d\pi_{\tilde{p}})^{-1}(V) \in T_{\tilde{p}}\tilde{M}$.

Define $\tilde{\gamma}(t) = \exp_{\tilde{p}}(t\tilde{V})$. Because local isometries map geodesics to geodesics, $\pi \circ \tilde{\gamma}$ is a geodesic in M with initial position $\pi(\tilde{p}) = p$ and initial velocity $d\pi_{\tilde{p}}(\tilde{V}) = V$. By the uniqueness of geodesics with given initial conditions, we must have $\pi \circ \tilde{\gamma} = \gamma$. □

Proof of the Theorem. **1. M is complete:** Choose any $p \in \pi(\tilde{M})$. Let γ be a geodesic in M starting at p. Let $\tilde{\gamma}$ be its lifted geodesic in $\tilde{M}$ starting at some $\tilde{p} \in \pi^{-1}(p)$. Since $\tilde{M}$ is complete, $\tilde{\gamma}$ is defined for all $t \in \mathbb{R}$. Consequently, $\pi \circ \tilde{\gamma} = \gamma$ is defined for all $t \in \mathbb{R}$. Thus, M is geodesically complete.

2. π is surjective: Choose any $\tilde{p} \in \tilde{M}$ and let $p = \pi(\tilde{p})$. Let $q \in M$ be any other point ($q \neq p$). Since we proved M is complete, by the Hopf-Rinow theorem, there exists a minimizing unit-speed geodesic $\gamma : p \to q$.

Let $d = d(p, q)$, so $\gamma(d) = q$. Lift γ to a geodesic $\tilde{\gamma}$ in $\tilde{M}$ starting at $\tilde{p}$. Because $\tilde{M}$ is complete, $\tilde{\gamma}$ is defined for all t, including $t = d$. Let $\tilde{q} = \tilde{\gamma}(d)$. We then have:

$$q = \gamma(d(p, q)) = \pi \circ \tilde{\gamma}(d(p, q)) = \pi(\tilde{q})$$

Thus, $q \in \mathrm{Im}(\pi)$, proving that π is surjective. (The property of being a covering map then follows from standard arguments utilizing the fact that geodesics lift uniquely and exp is a local diffeomorphism). □